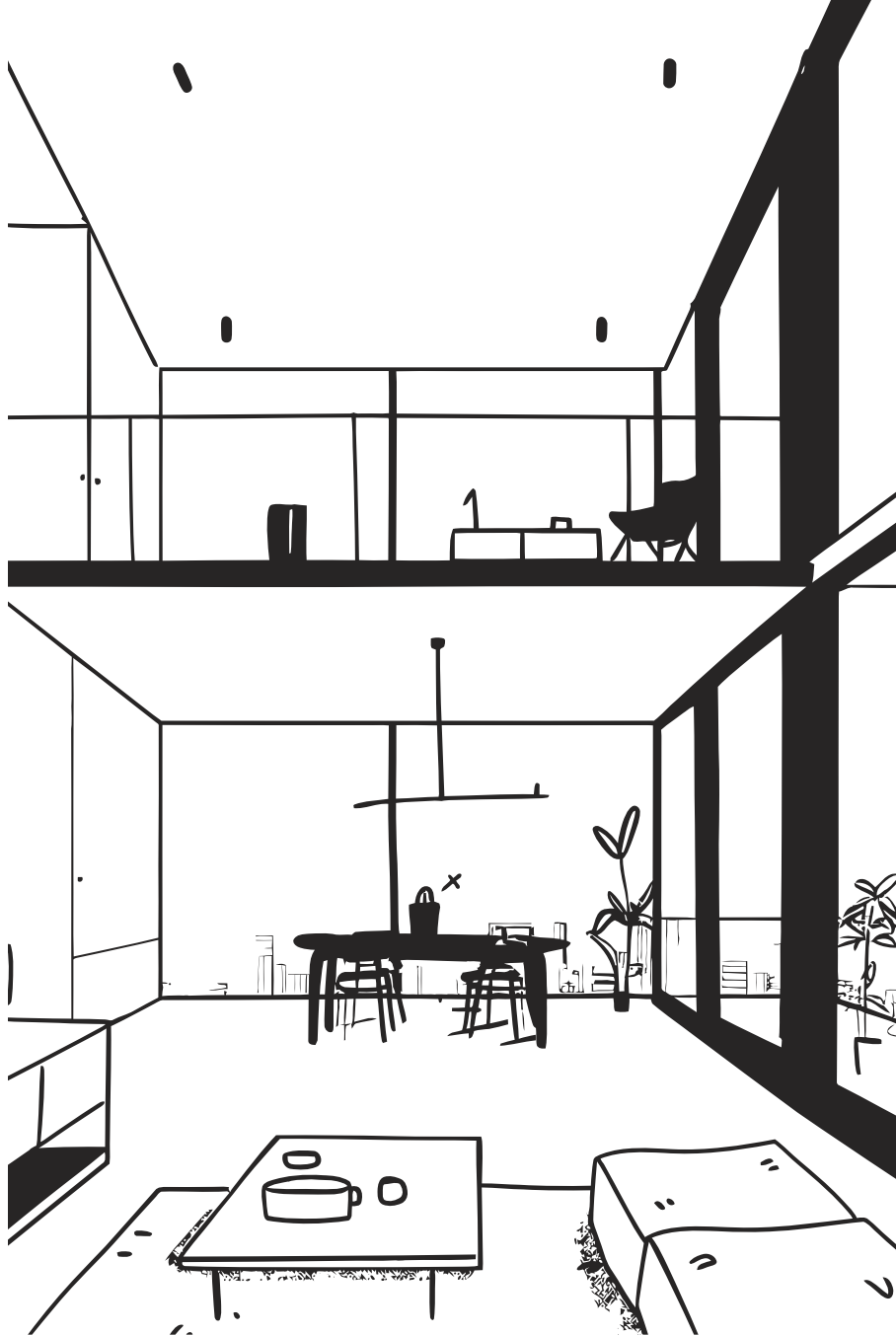
Permissioned vs Permissionless: the Ledger Choice

Not "blockchain yes or no" — but **which ledger, judged how.** The decision a business actually makes.

DRESCHER (2026), BLOCKCHAIN BASICS (2ND ED.), STEP 26



Two Ledger Models

PoW

Energy or computational work — Bitcoin

PoS

Economic stake — Ethereum

BFT

Voting among known validators — Fabric, Corda

PoA

Trusted authority nodes

How to Choose: Four Criteria

Trust

Who must you trust?

Throughput

How many transactions per second?

Governance

Who sets and enforces the rules?

Regulation

What must a supervisor see?

A Fabric transaction record contains

- **Channel** - the private sub-network, visible only to its members
- **Chaincode invoked** - the smart contract called
- **Read-set** - ledger keys read, with versions
- **Write-set** - keys and new values written
- **Endorsing organisations' signatures** - named, approved organisations

Fabric has **no native coin** - unlike a Bitcoin block, there is no coinbase reward.



Worked Example: Tokenised Sukuk

A Bahraini Islamic bank issues a tokenised sukuk and selects **Hyperledger Fabric**.

Why permissioned?

1 Governance

Only known, approved validators may participate

2 Throughput and privacy

High transaction volume, kept within a private channel

3 Regulation

CBB oversight requires an auditable, permissioned record

Why not public?

A permissionless chain fails the governance and regulation criteria - even though it is more open. Openness is not the right measure here.